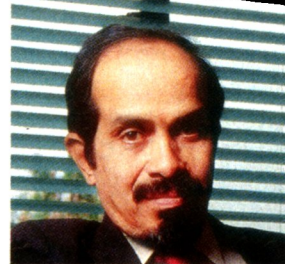


RODDAM NARASIMHA

Member, Scientific Advisory Council to the Prime Minister



Back to the Future

Instead of leading the world in R&D, India has become a blue-collar nation

During the first three decades after India's Independence, the economy grew only 3 per cent a year but the Science and Technology (S&T) infrastructure grew phenomenally. Decades later, when the economy is growing three times faster, the S&T infrastructure has not grown the way business and industry have. The public sector R&D departments (as well as national labs and premier research institutes), which used to harvest our Ph.Ds, have recently been unable to do so due to long-term job freezes. The private sector, with the exception of areas like pharma and biotech, has contributed little to the national R&D effort. The sudden boom

to a study, the number of publications from China and India were almost the same in 1997, but in the last decade, the proportional contribution of India to the world scientific community has for years just stood at approximately 2 per cent while that of Brazil, China and South Korea has gone up by factors of 3 to 10. Looking at Asia, India—ranked 22—is behind China and South Korea despite the country being the most cost-effective major source of open R&D in the world.

THE SOLUTIONS

► **Increase investment.** India's investment in R&D is about 1 per cent of the GDP compared to 2.6 per cent in South Korea and 1.3 per cent in China. The private sector in Korea invests twice as much as the government whereas in India such investment is less than a third. Here, most private industries do not need engineering or scientific Ph.Ds.

► **Arrest the decline of universities.** We have the potential to excel on the global front. We have one of the best space programmes operating on a shoe-string budget. More than 300 MNCs have their R&D centres in India leveraging Indian S&T talent but the flip-side is that traditional universities are in poor condition.

► **Make careers in S&T attractive for young Indians.** The present educational system should train the right kind of graduates in big numbers. Only 3,500 students from about five lakh candidates get selected for the IITs.

► **Create effective networks among academia, government and business.** This will promote technological innovation and remove the shackles on education and R&D. India should not remain a blue-collar S&T nation but must move up the innovation chain fast.

The scene in the country is still one of tall peaks over low averages. In 1947, Nehru had told the Indian Science Congress: "As soon as we get over our present troubles, there will be a flowering of science and other activities in India which will probably astonish the world." The question is: has the world been astonished by what India has done? Nehru liked to burn incense at the altar of science. How many of us have taken to that commitment with a religious devotion like his?

As told to Stephen David

700

is the number of Ph.Ds India has in engineering and technology compared to 9,000 in China in 2006.

HK RAJASHEKAR



THE PRESENT EDUCATIONAL SYSTEM NEEDS TO TRAIN THE RIGHT KIND OF GRADUATES IN BIG NUMBERS

and wealth creation in the information technology sector has drawn talent away from pure sciences.

As far back as 1958, Jawaharlal Nehru had Parliament pass a scientific policy resolution which said that "It was an inherent obligation of a great country like India with its tradition, scholarship and original thinking as well as its great cultural heritage to participate fully in modern science". In the long term, we need greater demand for S&T from private industry for India to have a sustainable S&T enterprise.

One widely accepted measure of a global R&D power is the number of globally recognised publications originating from the country. According